MATH 129: Big Data
HW8

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Problem 1: K-means

- 5,000 2D data points → need to implement K-means and determine appropriate number of clusters

Step 1: understanding the algorithm

K-means repeatedly performs 2 operations:

1 – Assignment: assign every point to its nearest cluster center

2 – Update: recalculate each cluster center as the mean of its assigned points

assignment rule: $c_i = \arg\min_j \|x_i - \mu_j\|_2$ (c_i)

update rule: $\mu_j = \frac{\sum_{i=1}^{m} 1\{c_i = j\} x_i}{\sum_{i=1}^{m} 1\{c_i = j\}}$

the objective is the sum of Euclidean distances: $J(c, \mu) = \sum_{i=1}^{m} \|x_i - \mu_{c_i}\|_2$

code implementation attached.

Problem: Extra Credit (Non-negative Matrix Factorization)

Goal: approximate the nonnegative Reuters TF-IDF matrix X by two smaller nonnegative matrices: $X \approx WH$. where W represents document-topic weights and H represents topic-word weights.

We'll minimize the squared Frobenius reconstruction error:

$$J(W, H) = \|X - WH\|_F^2$$

Using Lee and Seung's multiplicative updates:

$$H \leftarrow H \odot \frac{W^T X}{W^T W H + \epsilon}, \quad W \leftarrow W \odot \frac{X H^T}{W H H^T + \epsilon}$$

These updates preserve nonnegativity and, in exact arithmetic under the algorithm's assumptions, do not increase the objective.

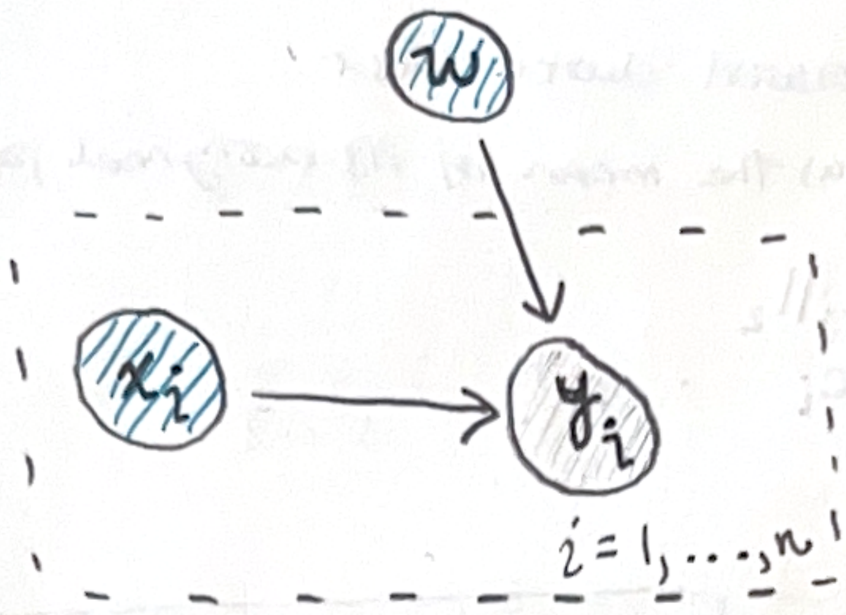
the code implementation is attached.

Problem Extra Credit: Bayesian Linear Regression

Given:

- generative model: $w \sim \mathcal{N}(0, V_0)$ and,
  - for $i = 1, \ldots, n$, $y_i | w, x_i \sim \mathcal{N}(w^T x_i, \sigma^2)$

(a) Express this model as a directed graphical model using Plate notation.



- The weight vector w is a global latent variable drawn from $\mathcal{N}(0, V_0)$. Each observation y_i, depends on w and the input x_i.
- The graphical model contains directed edges; $w \rightarrow y_i$ and $x_i \rightarrow y_i$.
- The variables x_i and y_i are enclosed in a plate indexed by $i = 1, \ldots, n$, while w remains outside the plate because it is shared across all observations

- The conditional likelihood factorizes as:

$$p(y_1, \ldots, y_n | w, D) = \prod_{i=1}^{n} \mathcal{N}(y_i | w^T x_i, \sigma^2)$$

(b) Is y_i independent of w?

In general, no, y_i is not independent of w. By the generative model,

$$p(y_i | w, x_i) = \mathcal{N}(y_i | w^T x_i, \sigma^2),$$

the conditional mean $w^T x_i$ depends on w. Therefore, knowing w generally changes the distribution of y_i and $y_i \not\perp w$.

- Conditioning on inputs does not block the direct path $w \rightarrow y_i$. In particular:

$$p(y_i | w, D) = \mathcal{N}(y_i | w^T x_i, \sigma^2).$$

- Marginalizing over the Gaussian prior on w gives:

$$p(y_i | D) = \mathcal{N}(y_i | 0, x_i^T V_0 x_i + \sigma^2).$$

- Since these distributions generally differ: $p(y_i | w, D) \neq p(y_i | D)$
- Hence, $y_i \not\perp w | D$.

(c) Is y_i independent of w given $D = \{x_i\}$?

- In general, no, y_i is not independent of w given $D = \{x_i\}$.

→ (the dependence vanishes only in degenerate cases, ie $x_i^T V_0 x_i = 0$.)